Multivariate Visual Analytics for Italian Regional Demographic and Healthcare Trends

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Abstract

Public demographic and healthcare datasets, such as those from ISTAT, offer rich spatiotemporal information but are difficult to explore due to high dimensionality. Current platforms primarily support univariate analysis, requiring manual toggling to identify correlations. This paper presents a Visual Analytics dashboard for exploring multivariate datasets with 32 indicators across 20 Italian regions (2002–2050). By integrating PCA and clustering with coordinated geographic and temporal views, the system detects complex regional patterns. A key contribution is the *Visual Triggering* paradigm: analytical algorithms like Feature Differentiation are activated via interaction (brushing), eliminating complex menus. Case studies on the North-South divide and healthcare sustainability demonstrate the system's effectiveness in hypothesis generation.

Project Repository: https://github.com/AntonioPagnotta/VA_project

1 Introduction

Public demographic and healthcare datasets, such as those provided by ISTAT and the Ministry of Health, offer crucial information for understanding socio-economic evolution but are difficult to explore due to high dimensionality. While these official sources represent a domain benchmark, existing platforms primarily support univariate analysis, forcing analysts to manually toggle between views to identify complex correlations. To bridge this gap, we propose a Visual Analytics dashboard tailored for a *Demographic Analyst* performing preliminary investigations.

Our solution integrates dimensionality reduction (PCA) and statistical profiling within a coordinated environment to support hypothesis generation. The analyzed dataset covers 32 indicators across 20 regions over 50 years (2002–2050), totaling approximately 32,000 observations.

This enables users to easily identify regional anomalies and structural patterns without disrupting their cognitive flow.

2 Related Work

This work builds upon three pillars: official statistical portals, academic GeoAnalytics, and narrative visualization. We highlight the advancements of our dashboard compared to these benchmarks.

2.1 Institutional Platforms (ISTAT & Ministry of Health)

Official databases , such as **ISTAT** [1] and the **Ministry of Health** [2], are the benchmark for data reliability but primarily support **univariate analysis**. Users must manually toggle between disparate views to identify correlations, leading to high cognitive load. Our solution is natively multivariate, integrating 32 indicators into a single coordinated environment to reveal cross-variable relationships without losing regional context.

2.2 Web-Enabled GeoAnalytics and Visual Guidance

Following **Web-Enabled GeoAnalytics** principles [3], we adopt a **Coordinated Multiple Views (CMV)** architecture. In the domain of demographic analysis, previous systems like **DemographicVis** [7] have successfully utilized techniques such as Parallel Sets and t-SNE clustering to link categorical attributes with user-generated content, proving the effectiveness of identifying user groups based on shared interests.

However, for high-dimensional quantitative data (> 30 variables), flow-based techniques often suffer from visual clutter. To mitigate this, we integrate **computational guidance** mechanisms. Similar to how the **STRive** system [6] utilizes Association Rule Mining to automatically direct analysts toward meaningful spatiotemporal patterns in complex datasets, our dashboard employs an automated *Feature Differentiation* algorithm (Top-K Ranking). This replaces manual line-tracing with semantic interpretability, highlighting the specific variables that make a selected region distinct.

2.3 The Gapminder Paradigm

Inspired by **Gapminder** [4], we adopt the **Bubble Chart** for temporal trend analysis. We enhance this model by linking it to the PCA clustering engine, enabling users to track not only movement along the axes but also the structural evolution of regions as they change cluster membership over time.

3 Data Abstraction and Preprocessing

3.1 Dataset Composition and Sources

The dataset integrates Italian demographic shifts with healthcare resource capacity and patient satisfaction at the **NUTS-2 level** (20 regions). The temporal dimension (2002–2050) combines historical records with predictive models. Data were synthesized from **ISTAT** (demographics and projections) and the **Ministry of Health** (bed and staff density). The resulting dataset comprises 32 quantitative indicators across 1,000 spatiotemporal units, totaling ≈32,000 observations.

3.2 Preprocessing and Feature Selection

To resolve heterogeneous units (percentages, scores 0–10, absolute counts), we applied **Z-score normalization**. A correlation analysis on historical data identified key drivers, such as the positive correlation between Staff Density and Longevity (+0.51) and the negative correlation between Aging and Satisfaction (−0.45). These insights guided feature selection, discarding irrelevant variables to streamline predictive modeling.

3.3 Predictive Methodology: Chained ARIMAX Modeling

Future projections up to 2050 were generated using a **chained forecasting architecture**:

- **Stage 1 (ARIMA):** Predicts structural resource evolution (supply side), such as Bed and Staff Density, based on historical trends.
- **Stage 2 (ARIMA):** Uses Stage 1 outputs as exogenous regressors to forecast human-centric outcomes (e.g., Healthcare Satisfaction Index, Healthy Life Expectancy).

It's interesting to notice thathe model highlights a future scenario where satisfaction metrics decline under the increasing pressure of an aging demographic, despite resource adjustments.

4 Visual Analytics Design

4.1 Design Overview and Layout

The dashboard is an integrated Single-Page Application designed according to Shneiderman's mantra: "Overview first, zoom and filter, then details-on-demand." To ensure simultaneous visibility and avoid vertical scrolling, the interface utilizes a fixed-height layout (`100vh`) with an asymmetric 40%–60% CSS grid.

The **left column (40%)** establishes context, housing the *Geospatial Map* for territorial anchoring and the *Cluster Chart* (dimensionality reduction) for structural similarity. This arrangement leverages the F-pattern reading habit to establish the geographic context first.

To ensure clarity and accessibility, the color schemes used in all visualizations are based on ColorBrewer scales [5], chosen for their perceptual uniformity and the ability to highlight patterns effectively. These color schemes are applied consistently to enhance visual differentiation without overwhelming the user, adhering to best practices for colorblind-friendly design.

The **right column (60%)** is dedicated to high-density detail, containing the *Bubble Chart* for multivariate correlations and the *Analytical Table* for feature differentiation.

Each visualization is encapsulated in a standardized card container with localized controls (dropdowns, sliders) to minimize cognitive load. The system supports two operational states: **Overview Mode** for static cross-sectional analysis and **Temporal Mode** for synchronized animation of historical and predictive trends (2002–2050), maintaining the user's mental map through consistent spatial arrangement.

4.2 Visual Encodings and Components

4.2.1 Geospatial Distribution (Choropleth Map)

The geographic view, implemented via `Leaflet.js` using NUTS-2 GeoJSON boundaries, serves as the spatial anchor for detecting territorial patterns. It employs a **double-state** visual encoding strategy:

- **Default State (Cluster-based):** Region fill encodes K-Means cluster membership using a qualitative 8-tone color scale to identify demographically homogeneous macro-regions.
- **Variable-specific State (Heatmap):** Upon selecting a specific indicator, the map transitions to a sequential orange scale (`#fff5eb` to `#7f2704`). Values are **globally normalized** ($[0, 1]$) relative to the 2002–2050

dataset extremes to ensure visual consistency during temporal animations. Missing data is rendered in neutral gray (`#e6e7eb`).

Interaction and Feedback Pre-attentive highlighting (border thickening) and tooltips provide details-on-demand during hover. A persistent selection state is activated via click, utilizing a **double-stroke technique** (3px black inner, 6px white outer border) to guarantee contrast against any background. This selection acts as a primary spatial filter, updating the Bubble Chart and Analytical Table. A dynamic legend discretizes the continuous scale into 7 quantitative bins for rapid value estimation.

4.2.2 Dimensionality Reduction & Clustering View

This component addresses the dataset's high dimensionality by projecting 32 indicators into a 2D space to reveal structural similarity patterns.

Analytical Pipelines The visualization is driven by two server-side processes:

- **PCA:** Projects multivariate profiles onto the first two principal components to maximize explained variance.
- **K-Means Clustering:** Groups regions into K homogeneous clusters. The system supports manual K selection (2–8) or an "Auto" mode via knee-point detection.

Visual Encodings and Interaction Implemented via *Chart.js*, the scatter plot maps latent coordinates to dimensionless axes. **Hue** encodes cluster IDs using a palette linked to the Choropleth Map for consistency. **Point geometry** dynamically reflects the selection state: unselected points (5px radius) expand (8px radius, 3px black border) to create a pop-out effect.

The core interaction is the **Visual Triggering** mechanism via a custom rectangular brush. Dragging a lasso around points identifies the selection and instantly executes a *Feature Differentiation Analysis*. This populates the Analytical Table to mathematically explain the similarity of the captured regions. In temporal mode, point positions are **linearly interpolated** (2002–2050), allowing analysts to observe regional trajectories and cluster migrations.

4.2.3 Bubble Chart (Variable Comparison)

Inspired by the Gapminder paradigm, this view investigates direct relationships between specific indicators through a trivariate scatter plot.

Visual Encodings The chart maps three quantitative variables to the X/Y axes and bubble **Size** (Area), the latter utilizing linear min-max normalization for temporal consistency. **Color** encodes cluster membership, ensuring visual linking and allowing users to verify if PCA clusters occupy distinct regions in the observed variable space.

Temporal Trails and Interaction A key feature is the management of the temporal dimension via **historical trails**. Selection triggers a trajectory composed of a dashed *connecting line*, discrete *historical dots* as temporal anchors, and the *current bubble*. Interaction is supported by three custom plugins:

- **Interactive Crosshair:** projects dashed guidelines to axes with exact value labels.
- **Semantic Tooltip:** provides details-on-demand with intelligent repositioning to avoid occlusion.
- **Year Watermark:** renders a semi-transparent label at the chart center to maintain temporal context during animations.

4.2.4 Top K Diverse Variables (Analytical Table)

While PCA identifies similarities, the *Analytical Table* provides semantic interpretability by offering an automated statistical explanation of the identified visual structures.

Feature Ranking Logic The table is driven by a dynamic pipeline triggered by user interaction (e.g., brushing). For each of the 32 indicators, the system calculates a discriminativity score based on the normalized difference of means:

$$Score = \frac{|\mu_{selected} - \mu_{rest}|}{\sigma_{global}}$$

Variables are ranked by this score, and only the Top K (user-adjustable, default 5) are displayed, effectively identifying the indicators that make a selection unique.

Visual Encoding and Interaction The component utilizes a dense **heatmap matrix** where columns represent the Top K variables and rows show selected regions alongside Cluster Medians for baseline comparison. Cell backgrounds use a sequential orange scale, with normalization occurring *per column* to highlight extreme values relative to each specific indicator's distribution.

The view supports two interaction states: **Auto Mode**, where the algorithm identifies statistically relevant variables, and **Manual Mode**, allowing analysts to toggle specific indicators for hypothesis confirmation.

4.3 Interaction Design and Visual Triggering

The dashboard utilizes the *Coordinated Multiple Views* (CMV) paradigm to transform the tool into an analytical reasoning environment. The architecture is based on three pillars: Brushing & Linking, Visual Triggering, and Temporal Navigation.

4.3.1 Coordinated Highlighting (Linking)

The system ensures cross-exploration context through a tight visual linking mechanism. Every selection (spatial or structural) propagates instantly across all views using **style uniformity** (3px black borders and increased size). In the Bubble Chart, a **spotlight effect** is achieved by reducing unselected observation opacity to 0.1, minimizing cognitive load while preserving global context.

4.3.2 Visual Triggering of Analytics

Advanced analytics are activated exclusively through direct manipulation, eliminating menus. The primary mechanism is **Brushing in the PCA space**:

- **Action:** Users draw a rectangular lasso to capture structurally similar regions.
- **Trigger & Computation:** Mouse release (mouseup) executes a real-time pipeline, calculating variance and z-scores ($Score = \frac{|\mu_{selected} - \mu_{rest}|}{\sigma_{global}}$) across all 32 variables.
- **Result:** The Analytical Table automatically reorders to show the most significant discriminants.

This loop (Action → Computation → Visualization) allows analysts to derive semantic insights directly from point geometry.

4.3.3 Temporal Navigation and Interpolation

The 2002–2050 temporal analysis is managed by an interactive player using **linear interpolation**. This avoids change blindness by visualizing the direction and speed of demographic shifts. Traceability is supported by **historical trails** in the Bubble Chart, which render dashed connecting lines and discrete historical anchors to reveal regional trajectories.

5 Analytics and Implementation

The system utilizes a modular **Client-Server** architecture to decouple analytical computation from real-time interaction, ensuring responsiveness during complex multivariate queries.

5.1 Technology Stack

The **Backend** is developed in **Python**, using **Pandas** for data manipulation and **Scikit-learn** for PCA and K-Means clustering. Communication is managed via **Flask** RESTful APIs. The **Frontend** is a Single Page Application (SPA) built with **Vanilla JavaScript** to maximize performance. Geospatial rendering is handled by **Leaflet.js**, while statistical visualizations (Scatter and Bubble Charts) use **Chart.js**.

5.2 System Architecture

The system adopts a **hybrid approach** combining pre-computation and real-time processing:

- **Pre-computation:** Baseline PCA projections for 2002–2050 are pre-calculated to facilitate fluid temporal navigation.
- **Real-Time Engine:** Triggered by interaction, the server re-executes K-Means for dynamic K selection and performs *Feature Differentiation* to identify discriminative variables for selected regions.
- **Synchronization:** The visualization layer captures user events (brushing/clicking) and queries the backend, ensuring all coordinated views remain synchronized with server-side models.

6 Case Studies and Insights

To demonstrate the system's effectiveness, we present a simulated workflow investigating structural disparities and temporal evolution across Italian macro-regions.

6.1 Intended User and Analysis Goals

The dashboard is designed for **Demographic Analysts** and **Policy Makers** conducting preliminary investigations on high-dimensional datasets. While official platforms often limit users to univariate views, our system supports:

- **Multivariate Exploration:** Understanding interactions between heterogeneous indicators (e.g., healthcare satisfaction and migration).
- **Hypothesis Generation:** Identifying outliers and unexpected patterns to formulate and validate analytical theories.
- **Spatiotemporal Integration:** Contextualizing phenomena geographically and temporally (2002–2050) within a single environment.

The workflow follows the **Overview-to-Detail** paradigm: the analyst establishes a structural overview via PCA and isolates clusters of interest to trigger in-depth statistical differentiation.

6.2 Insight: The North-South Divide and Growth Projections

The analyst applies the Overview-to-Detail paradigm to validate the hypothesis of persistent demographic polarization across Italian macro-regions.

Phases 1-2: Pattern Identification and Spatial Context The investigation begins with the **PCA Scatter Plot**, where Clusters 2 and 4 appear distinct. Brushing these anomalous groups triggers the **Analytical Table**, revealing *Net migration rate (internal)* and *Growth rate* as the primary discriminants (high z-scores). The **Choropleth Map**, updated via linking, identifies these regions as Southern Italy (excluding Lazio and Valle d'Aosta). Refined selection shows Southern regions occupying the bottom-left PCA quadrant, with *Life expectancy* and *Marriage rate* emerging as further critical variables.

Phase 3: Temporal Analysis and Prediction Using the **Bubble Chart** (X: internal migration, Y: female life expectancy at 65, Size: growth rate), the analyst observes Southern regions confined to a quadrant of negative values. Dynamic exploration (2002–2050) confirms the absence of a "catch-up" trend. Official forecasts suggest the *Growth rate* will further amplify existing inequalities, contradicting hypotheses of spontaneous demographic rebalancing.

To better make this insight studied, please refer to the Appendix 1 to observe this through the dashboard built

6.3 Insight: The Liguria Case — Migratory Rebalancing

Liguria serves as a case study for unique regional trajectories that deviate from general national trends.

Phases 1-2: Outlier Identification and Temporal Shift In the **PCA Scatter Plot** overview, Liguria appears as a significant outlier on the first principal component. The **Analytical Table** identifies the primary discriminants as extreme aging indices, with an exceptionally high ratio of elderly citizens. Transitioning from historical data (2002–2023) to future projections (2024–2050) reveals an unexpected shift: while historically isolated, Liguria shows a clear trajectory toward the main cluster in forecast periods, losing its outlier status. The **Top K Features** confirm this evolution, as discriminative power shifts from seniority indices toward migratory factors.

Phase 3: Validation and Rebalancing The **Bubble Chart** (X: Death rate, Y: Dependency ratio, Size: Total migration rate) validates this transition. Initially isolated in the high-aging/dependency quadrant, Liguria's bubble gradually moves into the main cloud as its size (migration rate) increases significantly. These findings, supported by the **Choropleth Map**, suggest that high immigration levels will act as a structural rebalancer, mitigating Liguria's historical demographic isolation and reintegrating its profile into the national average.

To gain a deeper understanding of this phenomenon, the reader is referred to the Appendix 2, where a direct comparison between the overall historical overview and the focused 2050 projection is presented.

6.4 Insight: Healthcare Sustainability Crisis

This case study correlates demographic shifts with healthcare performance to forecast the sustainability of care standards.

Phases 1-2: Correlation and System Warnings Bivariate analysis reveals a moderate negative correlation (-0.45) between the *Aging Index* and *Healthcare Satisfaction*. Simulations up to 2050 under constant resource supply predict a "System Crash," notably in *Sardegna*, where satisfaction is forecasted to drop below the critical 6.0 threshold. *Feature Differentiation* distinguishes biological outcomes from perceived quality: while *Staff Density* correlates strongly with *Longevity* ($+0.51$), *Satisfaction* is more closely tied to *Bed Density* and infrastructure capacity. This suggests that staff expansion protects health, while infrastructure modernization is required to maintain user satisfaction.

Phase 3 and Conclusion Cluster analysis identifies "Efficiency Gaps" in younger regions that exhibit low satisfaction despite low demographic pressure, indicating management inefficiencies. Forecasts show a steep, non-linear increase in required staff density to meet future needs. The analysis concludes that linear hiring is insufficient; an aggressive expansion of the Geriatric and Nursing workforce is mandatory to prevent the projected collapse of the healthcare system's perceived quality.

7 Conclusion

This work presented a Visual Analytics approach integrating dimensionality reduction with spatiotemporal views for the exploration of Italian demographic trends. The *Visual Triggering* paradigm enables seamless transitions between spatial, structural, and longitudinal perspectives, successfully addressing the challenge of high-dimensional data interpretation. Case studies on Liguria's trajectory and the Southern healthcare crisis demonstrate the system's efficacy in generating actionable insights. Future developments will focus on NUTS-3 provincial granularity to reveal local disparities currently flattened by aggregation. Additionally, the integration of "what-if" simulations will enhance the dashboard's utility for strategic policy-making.

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A Dashboard Insight

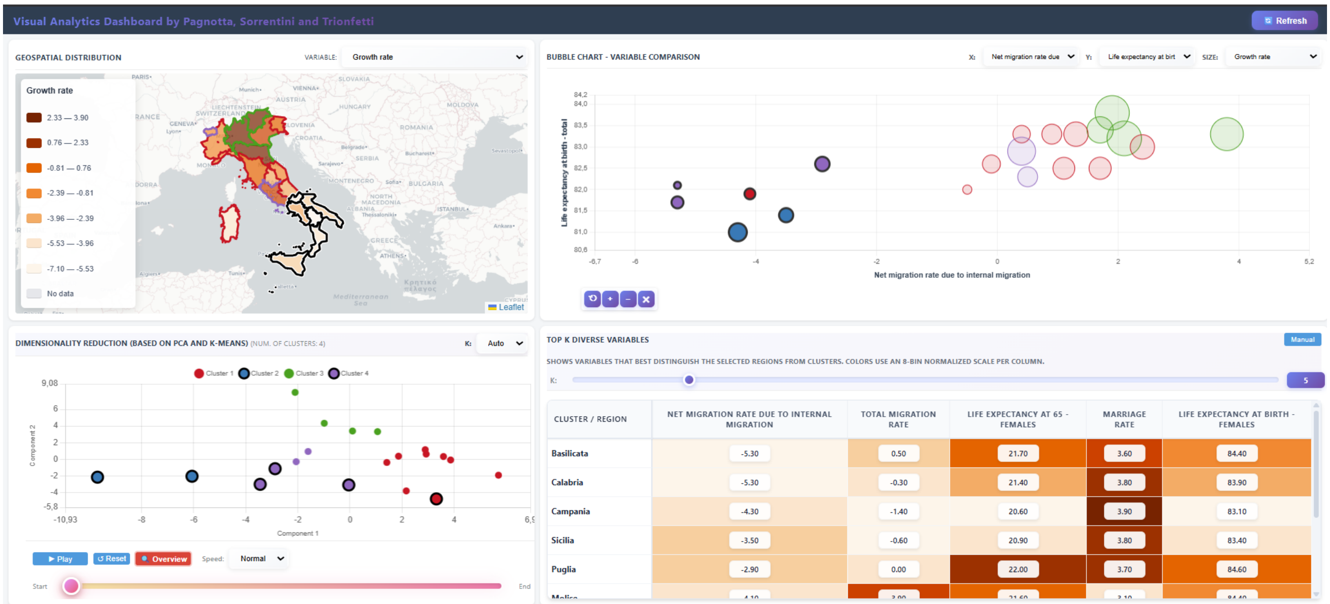

Figure 1: Interactive dashboard supporting the analysis of the North–South demographic divide.The PCA scatter plot highlights the separation of Southern Italian regions, which cluster in the lower-left quadrant and are characterized by negative internal migration and low growth rates.

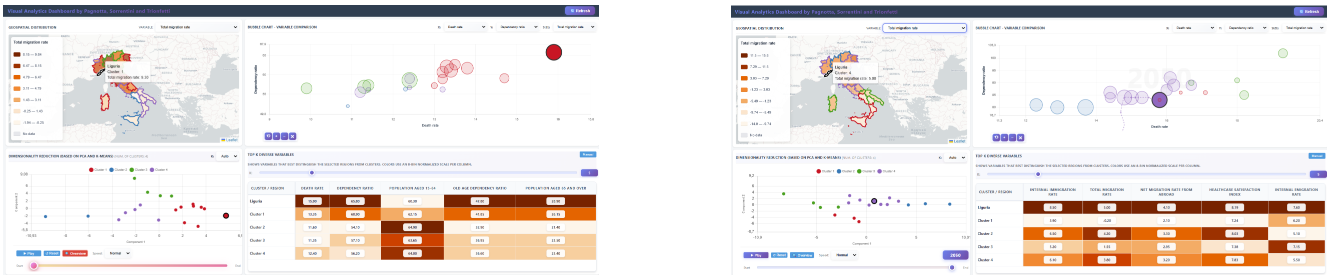

Figure 2: PCA scatter plots illustrating the evolving demographic position of Liguria. (*Left*) Overview based on overall data, where Liguria emerges as a clear outlier along the first principal component due to extreme aging and dependency indicators. (*Right*) Projection for 2050, showing Liguria converging toward the main cluster of regions, no longer isolated but embedded within the national point cloud, reflecting the increasing influence of migratory rebalancing processes.

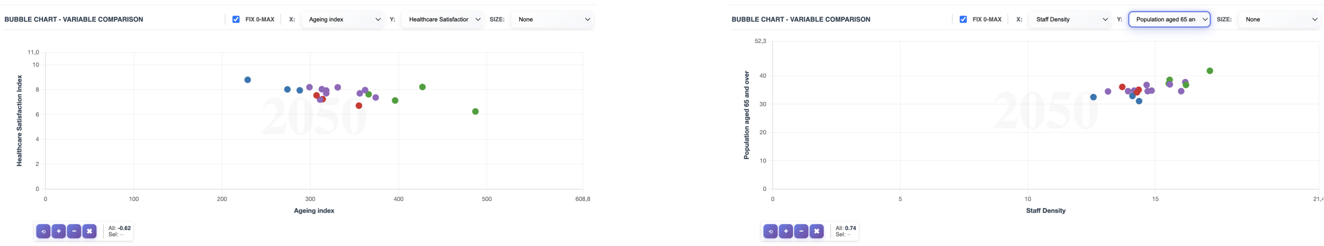

Figure 3: Bubble graph showing how the (*Left*) bivariate analysis reveals a strong negative correlation ($r = -0.62$) between the *Ageing Index* and *Healthcare Satisfaction*, projecting a 2050 "System Crash" where demographic pressure drives satisfaction levels below critical thresholds. (*Right*) A high positive correlation ($r = 0.74$) between *Staff Density* and the *Population aged 65+* highlights that sustaining future longevity requires an aggressive, non-linear expansion of the healthcare workforce rather than traditional linear hiring.